

Number System

Decimal
Number system

Numbers with
base 10

EX- $(102)_{10}, (24,579)_{10}, \dots$

Binary
Number system

Numbers with
base 2

$(1011)_2, (101,0011)_2, \dots$

Octal
Number system

Numbers with
base 8

$(245)_8, (212,57)_8, \dots$

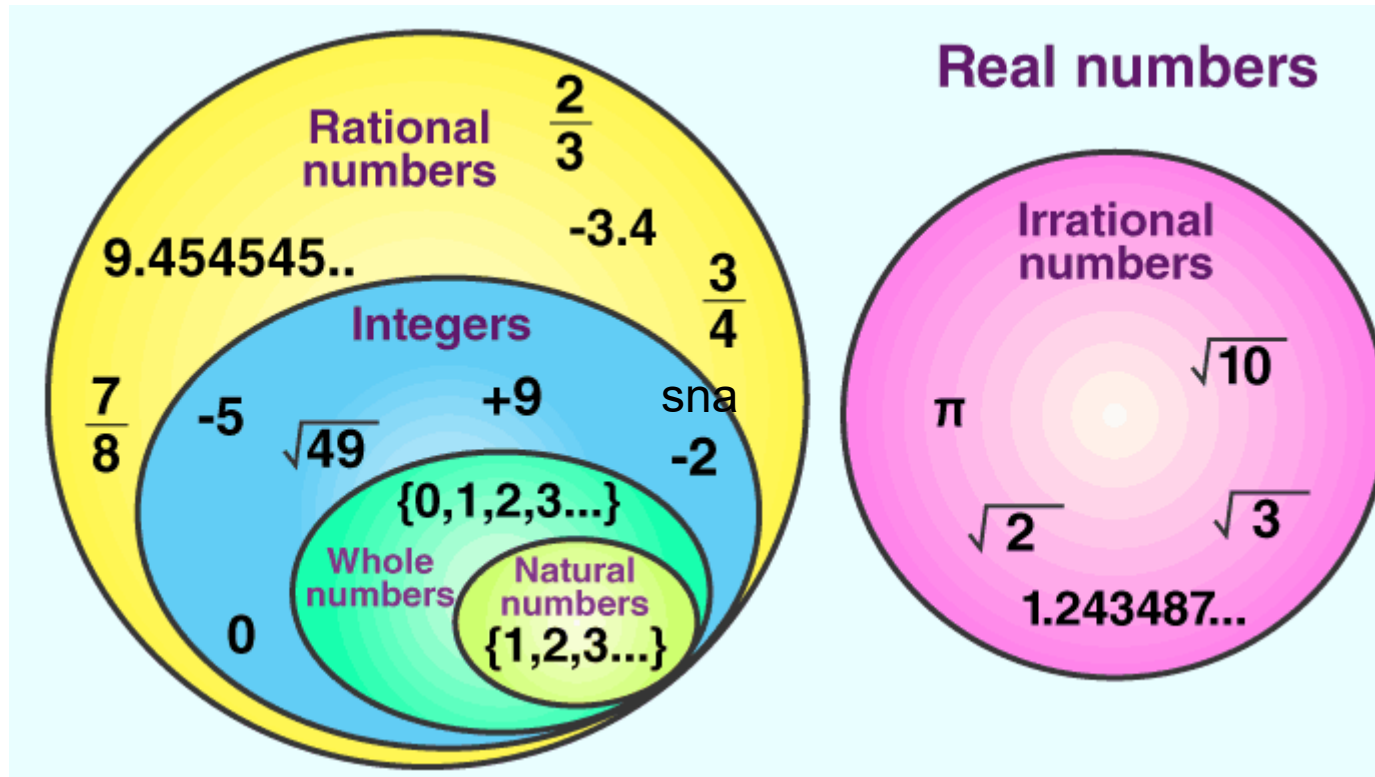
Hexadecimal
Number system

Numbers with
base 16

$(2B)_{16}, (D3F)_{16}, \dots$

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Briefing on Numbers



Number Quiz types of numbers Quizz
- Search



What is a Number System?

- A way to express or represent numbers using digits or symbols e.g. 123, ABCD, 12DE.
- A number system has a base (called **Radix** or **Base**).
- Radix indicates the number of digits supported by the number systems.
- **Common Number systems:**
 - **Decimal** (radix = 10) => digits 0 to 9
 - **Binary** (radix = 2) => digits 0 and 1
 - **Octal** (radix = 8) => digits 0 to 7
 - **Hexa-Decimal** (radix=16) => digits 0 to 9, symbols: A to F
- A number system radix base can be any value > 1 (as long as digits and symbols are well defined for it)

Representation: How do I know a number belongs to which number system?

- As the **decimal number system** is prevalent and most of us are comfortable with it, it is written without radix or base as it is understood.
- All other number systems follow a convention of showing the radix or base as subscript after the number
e.g. $(126)_8$ 28 or $(28)_{10}$ $(1FFF)_{16}$

Conversion between number systems: Can a number from one number system be converted to and from another Number System ? **Yes**

Binary Number System Representation (BCD)

- As a single memory cell can store one byte (8 bits) or information, binary numbers are generally written with additional zero bits if it is less than 8 bits. e.g.

$0 \Rightarrow (00000000)_2$, $7 \Rightarrow (00000111)_2$, $14 \Rightarrow (00001110)_2$

- The rightmost bit of a binary number is called the **LSB** (Least Significant Bit) and the leftmost bit is called **MSB** (Most significant bit). E.g,

0001100**1** (**LSB**) (**MSB**) **0**0011001

- A byte can further be grouped into 3 bits (for octal number) or 4 bits (for hexa decimal number). This simplifies the conversion between octal/hex-decimal to binary. e.g.

$(123)_8 \Leftrightarrow (001\ 010\ 111)_2$ $(9D2)_{16} \Leftrightarrow (1001\ 1101\ 0010)_2$

- Nibble** : A byte (8 bits) can be grouped into two 4bits numbers. Each 4-bit number is called a Nibble. e.g.

- $01010010 \Rightarrow 0101\ 0010$ (Nibbles can be easily mapped to hexadecimal)

- Converting binary to octal (grouping 3 bits from the LSB) e.g.

$(1011001001)_2 \Rightarrow (00)1\ 011\ 001\ 001 \Rightarrow (1311)_8$

- Converting binary to hexadecimal (grouping 4 bits from the LSB)

Number Conversion Table

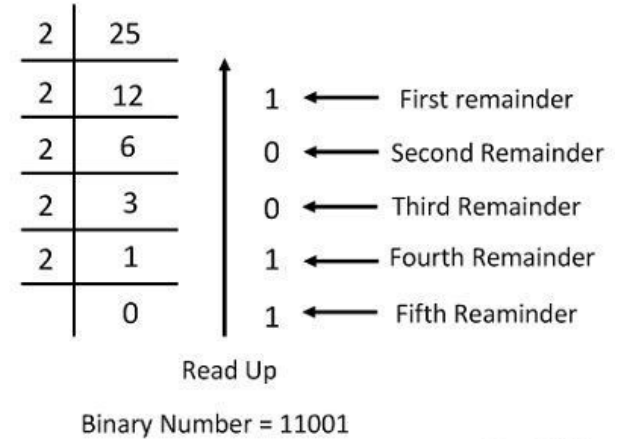
Decimal Numbers	Binary Numbers	Octal Numbers	Hexa-decimal Numbers
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	-	8
9	1001	-	9
10	1010	-	A
11	1011	-	B
12	1100	-	C
13	1101	-	D
14	1110	-	E
15	1111	-	F

Conversion Techniques

Double Dabble : Convert Decimal to any number system

Steps:

- Divide the number with the base of target number system (example shown is converting decimal to binary, hence dividing by 2)
- Write down the remainder as shown (next to the value after division)
- Keep dividing and writing down remainder until remainder is less than the base



Circuit Globe

Base factorisation: Convert any number system to Decimal

Steps:

- Divide the number with the base of target number system (example shown is converting decimal to binary, hence dividing by 2)
- Write down the remainder as shown (next to the value after division)
- Keep dividing and writing down remainder until remainder is less than the base

$$\begin{array}{c} 1 \quad 0 \quad 1 \quad 1 \quad 0 \\ \swarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \searrow \\ (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) \\ \downarrow \\ 16 + 0 + 4 + 2 + 0 \end{array}$$

Conversion Exercise

59 => binary

557 => binary

876 => octal

1876 => hexadecimal

$(11010111101)_2 \Rightarrow$ Decimal

$(1110110101011) =$ Octal

$(11100000001111)_2 \Rightarrow$ Hexa Decimal

$(1001110011101101)_2 \Rightarrow$ Hexa Decimal

$(561)_8 \Rightarrow$ Decimal

$(150)_8 \Rightarrow$ Decimal

$(15DF)_{16} \Rightarrow$ Decimal

$(FF)_{16} \Rightarrow$ Decimal

$(ABC)_{16} \Rightarrow$ Octal

$(752)_8 \Rightarrow$ Hexa Decimal

Remainder			
$2 \overline{)59}$	1	LSB	↑ Read up
$2 \overline{)29}$	1		
$2 \overline{)14}$	0		
$2 \overline{)7}$	1		
$2 \overline{)3}$	1		
$2 \overline{)1}$	1		
$2 \overline{)0}$	0	MSB	

A	B	C ₁₆
1010	1011	1100 ₂

101	010	111	100 ₂
5	2	7	4 ₈

Negative numbers in Binary number system

- **Sign Magnitude**

- Number of bits is fixed (8-bit signed, 16-bit signed, etc.)
- In this approach, the MSB indicates the sign (0 for plus, 1 for minus) e.g. :

8-bit signed $\Rightarrow (10000111)_2 \Rightarrow 1(0000111)_2 \Rightarrow -7$

16-bit signed $\Rightarrow (0000000000000111)_2 \Rightarrow 0(0000000000000111)_2 \Rightarrow +7$

- **2's Complement Method**

- Number of bits is fixed (8-bit , 16-bit , etc.)
- In this approach, the MSB indicates the sign (0 for plus, 1 for minus)
- A positive number can be converted to its negative by following method:

Find One's compliment of the number and add 1.

One's compliment is obtained by changing every 0 bit to 1 and 1 to 0

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-
- 8-bit system $\Rightarrow (00000111)_2 \Rightarrow +7$
- 1's compliment $\Rightarrow (11111000)_2$

- Add 1 $\Rightarrow (\quad \quad \quad 1)_2$
- 2's compliment $\Rightarrow (11111001)_2 \Rightarrow -7$

Convert back:

- 1's compliment $\Rightarrow (00000110)_2$
- Add 1 $\Rightarrow (\quad \quad \quad 1)_2$
- 2's compliment $\Rightarrow (00000111)_2 \Rightarrow +7$

teachoo

Binary Addition Table

Addition		Result	Carry
0 + 0	=	0	0
0 + 1	=	1	0
1 + 0	=	1	0
1 + 1	=	0	1

1	1	1	
1	1	0	1
+	1	1	1
1	0	1	0
0			

Subtraction in Binary number system

- **Subtraction using Borrowing method**
- In this method, subtraction is performed as per the borrowing table.
- **Subtraction using Using 2's compliment (Signed Numbers)**
- In this method, number to be subtracted is converted to its **2's compliment** form (which makes it negative) and added to the number from which it need to be subtracted. e.g.

Subtraction : $(00000111)_2 - (00000110)_2$

2's complement of $(00000110)_2 \Rightarrow (11111010)_2$

Addition $\Rightarrow (00000111)_2$

+ $(11111010)_2$

 $(00000001)_2$

Inputs		Outputs	
A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

BINARY SUBTRACTION

$\begin{array}{r} 1010 \\ - 101 \\ \hline 101 \end{array}$	$\begin{array}{r} 1100 \\ - 1011 \\ \hline 1 \end{array}$	$\begin{array}{r} 110000 \\ - 11110 \\ \hline 10010 \end{array}$
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Subtraction Exercise

Subtract using borrowing method

$$(1110101011101)_2 - (11111011111)_2$$

$$(1011111101)_2 - (1011001)_2$$

$$(100011)_2 - (101)_2$$

A diagram illustrating binary subtraction using the borrowing method. The top number is 11000 and the bottom number is 111. Red diagonal lines indicate borrowing from the 1s, 10s, and 100s places. Blue diagonal lines indicate the borrowing chain from the 1000s place. The result is 10001.

$$\begin{array}{r} 11000 \\ - 111 \\ \hline 10001 \end{array}$$

Subtract using 2's complement method

$$(01011101)_2 - (10010)_2$$

$$(11011111)_2 - (10111)_2$$

$$(10000001)_2 - (11000)_2$$

A diagram illustrating binary subtraction using the 2's complement method. It shows the addition of 1000 and 0110 to get 1110. This result is then complemented to 0010, and a minus sign is added to the final result.

$$\begin{array}{r} 1000 \\ + 0110 \\ \hline 1110 \end{array}$$

2's Complement

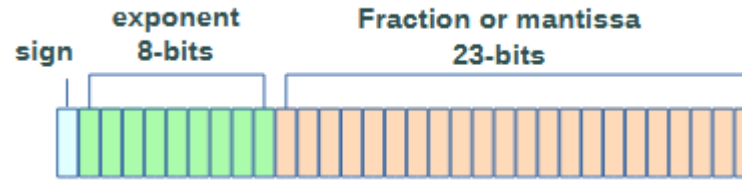
(-) 0010 ← True difference

Put minus sign

Real or Scientific Number Representation

(Fraction X Base^{Exponent})

How are real numbers represented in binary system?



IEEE 754 - 32-bit (Single Precision) number representation in binary

- MSB - Used to indicate Sign (positive or negative)
Exponent - 8 bits $2^{(E-127)} \Rightarrow \min=2^{-126} \max:2^{127}$
Fraction - 2^{23} (max value = 83,88,608)

Example: 33.33

$$(33)_{10} = (100001)_2$$

$$33.33 \Rightarrow 100001.100001 \Rightarrow 1.00001100001 \times 2^5$$

$$\text{Exponent} = 5 \quad \text{Mantissa} = 100001100001$$

$$\text{Biased Exponent} = 5 + 127 = 132 \Rightarrow 10000100$$

$$\text{Full Mantissa (23 bits)} = 10000110000100000000000$$

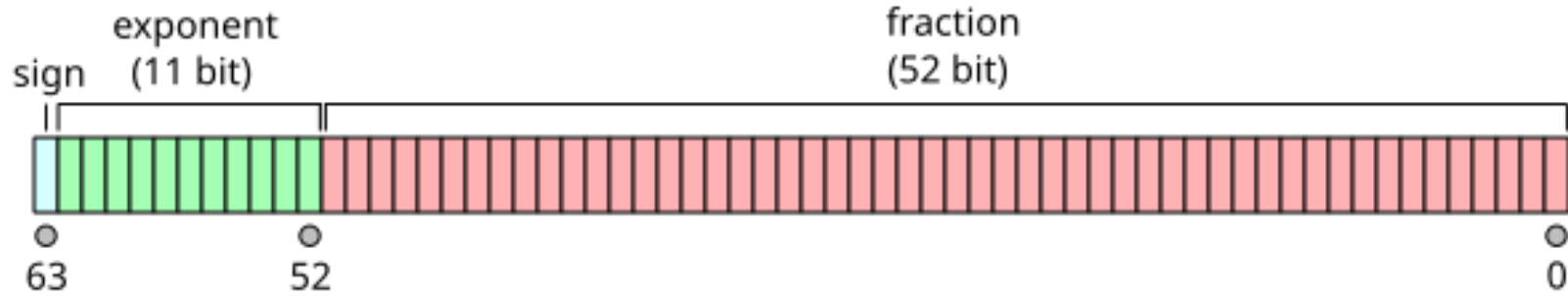
$$\text{Sign} = 0$$

$$\begin{aligned} 33.33 &\Rightarrow (0 \ 10000100 \ 10000110000100000000000)_2 \\ &\Rightarrow (42430800)_{16} \end{aligned}$$

In memory
representation

00000000
00001000
01000011
01000010

How are real numbers represented in binary system?



IEEE 754 A 64-bit (double precision) number representation in binary

MSB - Used to indicate Sign (positive or negative)
Exponent - 11 bits $2^{(E-1023)} \Rightarrow \text{min}=2^{-1022} \text{ max}:2^{1023}$
Fraction - 2^{52} (max value = 4,503,599,627,370,496)
Example: 33.33

$$33.33 \Rightarrow 100001.100001 \Rightarrow 1.00001100001 \times 2^5$$

Exponent = 5 Mantissa = 100001100001

$$\text{Biased Exponent} = 5 + 1023 = 1028 \Rightarrow (100000000100)_2$$

Full Mantissa (52 bits) = 100001100001000

Sign = 0

[illegible]

In memory representation

00000000
00000000
00000000
00000000
00000000
01100001
01001000
0100000

³Knowledge Check (Number System)

1) How many memory locations can be address with 10 wires?

[a] 1024 [b] 1000 [c] 1KB [d] 1Kb

2) A binary number can be converted to hexa decimal number by grouping how many bits of the binary digits.

[a] Nibble [b] 4 bits starting from LSB [c] 4 bits starting from MSB

3) Integer numbers belonging to any number system can be converted to Decimal using:

[a] Base factorization [b] Double dabble method

4) Integer numbers belonging to Decimal number system can be converted to any number system using:

[a] Base factorization [b] Double dabble method

5) What is a radix in number system?

[a] Base [b] number of digits and symbols in the number system

6) Are all these numbers equal ? 10, (A)₁₆, (12)₈, (1010)₂

[a] True [b] False

7) Choose the negative numbers from below signed binary numbers.

[a] 11110001 [b] 00011100 [c] 11000011 [d] 01110101

8) What is the 2's compliment of 0000 ?

[a] 10000 [b] 0000 [c] 1111 [d] 00001

9) What will the output of adding a number and its 2's compliment ?

[a] 0 [b] negative number [c] positive number

10) A decimal number needs ow many bytes in memory?

[a] 8 [b] 16 [c] 32 [d] 64